A Major Project Report
(Phase-I)

On

[PT137] Ms. Shalini Singh — Intelligent Credit Risk Assessment using Explainable AI

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

By

Kushagra Sharma (2410031018)

Nikhil Verma (2410031015)

Taufique Raza (2410031042)

Shams Shahzad (2410030720)

Under the guidance of Ms. Shalini Singh

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA

SCHOOL OF COMPUTER SCIENCE AND ENGG.

IILM UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**[PT137] Ms. Shalini Singh — Intelligent Credit Risk Assessment using Explainable AI**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Mr. Yash Garg**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

Kushagra Sharma (2410031018)
Nikhil Verma (2410031015)
Taufique Raza (2410031042)
Shams Shahzad (2410030720)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name and Details of Guide

Name: Ms. Shalini Singh Shalini., 27/8

Designation: Assistant Professor

School: School of Computing Science and Engineering

IILM University, Greater Noida

Table of Contents

SL. No	Section	Page no.
	Abstract	1
1	Problem Definition	2
2	Literature Survey	3
3	Objectives	4
4	Scope	5
5	Feasibility Study	6
	References	7
	Plagiarism Report	8

Intelligent Credit Risk Assessment using Explainable AI – Phase I: Project Initiation

Abstract

The process of credit risk assessment is crucial in controlling financial risk and impacts lending, default rate and stability. While modern machine learning models are superior at predicting outcomes compared to traditional statistical scoring models, their effectiveness comes at the cost of transparency, fair lending regulations and algorithmic bias due to “black box” nature of the models.

The solution proposed is called Intelligent Credit Risk Assessment System using Explainable AI (XAI).

The system is employing modern machine learning technologies i.e. LightGBM, Random Forest and XGBoost, which trained on the massive amount of historical credit data are capable of predicting the chance of loan default.

To explain the outcome of the trial, the methods of modern zero explanation technologies called SHAP and LIME are being applied. The resulting system is characterized by enough accuracy on the one hand and explanations that are understable by human being on the other hand, respecting all the regulations such as ECOA and GDPR.

Problem Definition

The problem of financial institutions is one that is relevant in the current world. In most situations, financial institutions use sophisticated machine learning systems to calculate the number of defaulted borrowers with great accuracy; however, there is an issue of the "black box" nature of these algorithms. The use of such solutions brings about challenges in terms of the regulations, operation of financial institutions, and ethical aspects of the work process. Automated lending could not provide the reason behind its decisions to clients; for example, increasing the number of approved loans; as a result, it's obvious that the existing procedures do not comply with the principles of fair lending, which can be illustrated by the examples of ECOA and GDPR.

In our work we are going to tackle the issue of the implementation of the AI system that will assess the risks in a transparent way. This is the goal of our paper, which implies the need for the creation of the explainable AI system as the background of conducting the required operations. We are going to create a theoretical model that will include approaches that provide the necessary explanation, as well as technologies such as SHAP and LIME.

Literature Survey

Credit risk evaluation has undergone three main stages:

1. Traditional Statistical Measurement

Initially used techniques were simple linear models like Logistic Regression or Linear Discriminant Analysis. These models were easy to interpret, but their linear nature limits their ability to handle complex and big data sets, which makes their predictions less precise.

2. Sophisticated Machine Learning Models

To increase accuracy, the industry started using ensemble methods, including XGBoost, LightGBM, or Random Forest. Despite being much better at predicting defaults, these algorithms are difficult to interpret because of their elaborate structure (the "black box" problem).

3. Introduction of Explainable Artificial Intelligence (XAI)

Contemporary approaches employ post-hoc XAI technologies to achieve a balance between high accuracy and regulatory compliance.

SHAP (Shapley Additive exPlanations): Applies cooperative game theory to explain the model globally (by providing the ranking of features based on their importance within the whole data set).

LIME (Local Interpretable Model-agnostic Explanations): Offers explanations of individual outcomes via local approximation of the complex model around the specific client.

The use of ensemble techniques in combination with SHAP and LIME allows organizations to achieve desired predictive results and at the same time conform to regulations like GDPR's "Right to Explanation."

Objectives

The main goal of the project is to design a credit scoring system that is accurate, reliable, and follows regulations while doing away with black-boxstyle machine learning systems. These are the objectives of the project:

1. Data Preparation: The approach includes gathering and processing financial object datasets, which will involve the process of treating missing entries, encoding categoricals, and finding key features which impact the risk assessment.

2. Model Development: The model will soon utilize various machine learning methods such as XGBoost, LightGBM, and Random Forest, which will allow assigning risky/non-risky status of loan applicants.

3. XAI Framework: The use of SHAP and LIME techniques will help in determining global feature importance metrics and how decisions will be made.

4. Model Performance and Compliance Assessment: The performance of the model will be assessed using the relevant metrics (ROC-AUC, true positive rate, false positive rate, and F1 score) conformance to lending legislation (i.e., GDPR and ECOA, etc.).

5. Real-time Dashboards: The development of an online application, which makes simple for credit agents to input applicant data and get the up-to-date risk metrics together with the corresponding visuals, will be done.

Scope

The project has the goal to create an Explainable AI (XAI) framework designed for intelligent credit risk assessment. It can be divided into the following stages:

1. Data Preparation: This stage involves manipulating financial datasets. The goal in this stage is data cleaning and transformation and the technique of missing value treatment and addressing class imbalance.

2. Model Training: The goal of this stage is to design a supervised machine learning model aimed at determining the probability of applicants' defaults.

3. Explainability Pipeline: The possibilities of SHAP and LIME will be used for the mapping of these features and the explanation creation for each decision taken.

4. Platform Development: In this stage a web-based application will be created. Streamlit or Flask will be used as the basis for making the application interactive to enter the applicant's data and see the classification results.

5. Effectiveness Evaluation: This stage involves the checking of the compliance with classification metrics and evaluation of the algorithm's performance.

Feasibility Study

The feasibility study analyzes the implementation feasibility, validity, and sustainability of the Intelligent Credit Risk Assessment System proposed. The analysis can be divided into four main aspects.

1. Technical Feasibility.

Equipment and tools: The system is using conventional toolkit with open-source libraries in Python, such as scikit-learn, XGBoost, SHAP, LIME, Streamlit, etc., thus not requiring any cutting-edge technologies.

Model Application: The system is using pre-trained ensemble algorithms and post-hoc interpretability methods that are easy to describe. Thus, the process of creating a data pipeline, training the model, and developing a dashboard is trivial.

2. Operational Feasibility.

Functionality: The interactive web-based dashboard provides graphical displays (for example, SHAP summarized charts and LIME local force plots) to credit regulators and loan officers instead of raw text.

Decision-making: Automated justification of attributes provides an explanation and understanding of the nature of the ranked loans to clients who have no technical knowledge.

3. Economic Feasibility

Cost: The method is appropriate since no costs to software development are incurred as only free tools and information, available in the public domain, is used.

Business Value: Ability to make more efficient predictions along with clear explanations of the reasons ensures effective use of the method and contributed to significant reduction of bad debts.

References

Academic Literature & Foundational Papers

- **Explainable AI (SHAP): Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS 2017)* (Vol. 30, pp. 4765–4774).**

- **Explainable AI (LIME): Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1135–1144).**

- **Ensemble Learning (XGBoost): Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794).**

- **Ensemble Learning (LightGBM): Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS 2017)* (Vol. 30, pp. 3146–3154).**

- **Ensemble Learning (Random Forest): Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.**

- **XAI in Credit Risk: Bussmann, N., Giudici, P., Marinelli, D., & Papenbrock, J. (2021). Explainable AI in fintech risk management. *Frontiers in Artificial Intelligence*, 3, 580124.**

Public Credit Risk Datasets & Portals

- **Home Credit Default Risk (Kaggle): Multi-table real-world dataset containing application details, credit bureau data, and previous loan behaviors.**

- **German Credit Data (UCI Machine Learning Repository): Standard financial benchmark with 1,000 credit records evaluating good vs. bad credit risks.**

- **Give Me Some Credit (Kaggle): Financial challenge dataset designed to predict the probability that somebody will experience financial distress in the next two years.**

- **LendingClub Loan Data: Historical marketplace lending dataset with credit grades, payment history, and default outcomes**

Open-Source Software & Toolkits

- **SHAP (SHapley Additive exPlanations): Python documentation and repository for tree explainers and local force plots.**

- **LIME (Local Interpretable Model-agnostic Explanations): Python package for perturbation-based local surrogate models.**

- **Scikit-learn: Data preprocessing, imputation, standard classification baselines, and performance metrics (ROC-AUC, F1-score).**

- **Streamlit: Python framework for building interactive data science dashboards and real-time model interfaces.**

Plagiarism Report

Dupli Checker

Date: 19-08-2026

Plagiarism Scan Report

4% Plagiarism	4% Exact Match	96% Unique	Words 1000
	0% Partial Match		Characters 6775
			Sentences 125
			Paragraphs 63
			Read Time 8 minute(s)
			Speak Time 10 minute(s)

Content Checked For Plagiarism

Intelligent Credit Risk Assessment using Explainable AI – Phase I: Project Initiation

Abstract

The process of credit risk assessment is crucial in controlling financial risk and impacts lending, default rate and stability. While modern machine learning models are superior at predicting outcomes compared to traditional statistical scoring models, their effectiveness comes at the cost of transparency, fair lending regulations and algorithmic bias due to "black box" nature of the models.

The solution proposed is called Intelligent Credit Risk Assessment System using Explainable AI (XAI).

The system is employing modern machine learning technologies i.e. LightGBM, Random Forest and XGBoost, which trained on the massive amount of historical credit data are capable of predicting the chance of loan default.

To explain the outcome of the trial, the methods of modern zero explanation technologies called SHAP and LIME are being applied. The resulting system is characterized by enough accuracy on the one hand and explanations that are understable by human being on the other hand, respecting all the regulations such as ECOA and GDPR.

Shahi
27/8